

From the Calibrated Model to the Estimated Model: Modifications Required Beyond Subsection 4.2

August 10, 2026

Abstract

This note catalogs every modification made to the theoretical model in `ShotgunPCestCovid.tex` – as fully specified through Section 1 (Introduction), Section 2 (Budget Constraints), Section 3 (The Model), and Subsections 4.1–4.2 (Model Summary and Calibration) – in order to arrive at the model actually taken to the data, `mmt8b0ccBin3_covid_bT.baa.mod`, and its Bayesian estimation results (current round: reserve-equation fix, diffuse b_π, b_y priors, `hessian` jumping covariance, $\bar{\tau} = 0.14$, $\chi = 0.5$; see `mmt8b0ccBin3_covid_bT.baa.epsL.summary.tex` for the full methodology and results, including the superseded tight-prior “gen1” round retained there for comparison). It is intended as a working reference for rewriting the paper’s Subsection 4.3 onward, which currently continues with the calibrated-model determinacy analysis and an older, no-longer-accurate Section 5 estimation draft (the latter lacks the debt-to-GDP observable, the Baa-based loan rate, the COVID labor-supply shock, a genuine OccBin ZLB likelihood, and uses a different fiscal calibration than the results actually obtained). No existing files are modified by this note; it is a planning document only.

Contents

1	Scope and a frequency caveat	2
2	Structural equation modifications	2
2.1	Interest rate rule: output response, steady-state rate base, shock, and ZLB . . .	2
2.2	Tax rule: lagged debt, persistence, shock	2
2.3	Debt-issuance equation: linearization and simplification	3
2.4	Euler equation: marginal-utility reformulation (solution technique, not economics)	3
2.5	Labor-leisure condition: productivity in place of the wage, plus the COVID shock	4
2.6	New variable: debt-to-GDP link	4
2.7	Reserve equation: restored debt-state coupling (update)	4
3	New stochastic structure	5
4	Observable/measurement block (entirely new)	5
5	Calibration changes	6
6	Parameters moved from calibrated to estimated	7
7	Estimation methodology (absent from Sections 1–4.2 by construction)	8

1 Scope and a frequency caveat

Sections 1–3 and Subsections 4.1–4.2 present the model at a calibration that does not specify an explicit sampling frequency, with illustrative annual-scale targets ($\Pi = 1.04$, $r = 1.07$, $\beta = 0.972$, etc.). The estimated model is explicitly *quarterly*, disciplined by a 2009Q1–2025Q4 U.S. sample, and every rate-like calibrated value is recalibrated accordingly – the same symbol (e.g. β) refers to a materially different numerical target between the two. This is not itemized separately below except where the reinterpretation also changes the model’s economic content (e.g. the ZLB), but should be kept in mind throughout: a side-by-side value comparison for shared symbols is only meaningful once both are understood as quarterly.

2 Structural equation modifications

2.1 Interest rate rule: output response, steady-state rate base, shock, and ZLB

Paper (eq. ??-equivalent, Section 2):

$$r_t^n = r_t^{\alpha_i} \left(\frac{\Pi_t}{\Pi} \right)^{b_\pi}.$$

Estimated model:

$$r_t^n = \max \left[1, (r_t^{\ell,ss})^{a_i} \left(\frac{\pi_t}{\pi^{ss}} \right)^{b_\pi} \left(\frac{y_t}{y^{ss}} \right)^{b_y} \exp(z_t^r) \right], \quad z_t^r = b_r z_{t-1}^r + \varepsilon_t^r.$$

Three changes: (i) an output-response term $(y_t/y^{ss})^{b_y}$ is added (no output term exists in the paper’s rule at all); (ii) the base rate is the *fixed calibrated steady-state* loan rate $r^{\ell,ss}$ rather than the time-varying market rate r_t (the paper’s r_t and loan rate r_t^ℓ coincide via the arbitrage condition $r_t^\ell = r_t$, but the estimated rule uses the calibrated anchor, not the contemporaneous value); (iii) a persistent monetary-policy shock z_t^r is added, entirely absent from the paper. Most substantively, (iv) a strict zero lower bound is imposed via a $\max[1, \cdot]$ operator, solved with Dynare’s OccBin piecewise-linear toolkit (Guerrieri and Iacoviello, 2015) – the paper’s calibrated model has no lower bound on r_t^n at all. Enforcing the ZLB genuinely (rather than carrying the constraint syntactically but disabling it in estimation, as an earlier draft of this file did) requires OccBin’s true piecewise-linear likelihood, which is itself nonsmooth and required a two-stage mode-finding procedure (Section 6 below).

2.2 Tax rule: lagged debt, persistence, shock

Paper (eq. ??):

$$\tau_t = \bar{\tau} \left(\frac{b_t}{(1 - \Omega)b_0^T} \right)^{\varphi_b}.$$

Estimated model:

$$\tau_t = \tau_{t-1}^{\rho_\tau} \bar{\tau}^{1-\rho_\tau} \left(\frac{b_{t-1}}{(1 - \omega)b_0} \right)^{\varphi_b} \exp(\varepsilon_t^\tau).$$

Three changes: the debt ratio is *lagged* (b_{t-1} , not b_t) – required so that b_t is a predetermined state variable, which matters for Blanchard-Kahn compatibility once b_t is genuinely endogenous

(see Section 2.3); an AR(1)-type persistence term $\tau_{t-1}^{\rho_\tau}$ is added (the paper’s rule is a purely static function of the debt ratio); and a stochastic innovation ε_t^τ is added (the paper’s rule is deterministic).

2.3 Debt-issuance equation: linearization and simplification

The paper actually contains *two* separate real budget-constraint identities (Section 3, “Equilibrium Household Budget Constraints”), not one: eq. (??), which solves for currency, $m_t = g_t - \tau_t + r_t^n n_t$, and eq. (??), which solves for reserves, $n_t = m_{t-1}/\Pi_t + b_{t-1}/\Pi_t - b_t/r_t$ – the latter is the one *including* the reserve stock n_t and the interest-on-reserves rate r_t^n (via m_t ’s own definition) explicitly, and is discussed separately in Subsection 2.7 below, since the estimated model modifies it too, in the opposite direction (restoring rather than dropping terms). The two identities combine into one consolidated government budget constraint. The Policy subsection then writes a *third*, separate equation – eq. (??) below – that rearranges this same consolidated identity to solve directly for b_t instead of n_t ; being a rearrangement of the same underlying identity, it too contains m_{t-1} and n_t/r_t^n terms. This subsection concerns only that third equation, the one solving for b_t .

Paper (eq. ??):

$$b_t = r_t [b_{t-1}\Pi_t^{-1} + g - \tau_t - m_{t-1}(1 - \Pi_t) + (r_t^n - 1)n_t].$$

Estimated model:

$$b_t = \frac{r^{\ell,ss}}{\pi^{ss}} b_{t-1} + g_t - \tau_t + b_0 \left(1 - \frac{r^{\ell,ss}}{\pi^{ss}} \right) - (g - \bar{r}).$$

This is the single largest structural departure from the paper. The time-varying ratio r_t/Π_t multiplying b_{t-1} is replaced by a *fixed* calibrated ratio $r^{\ell,ss}/\pi^{ss}$ (an extension of Assumption 2’s fixed-real-rate logic to the dynamic debt equation); the currency-growth term $-m_{t-1}(1 - \Pi_t)$ and the reserve-interest term $(r_t^n - 1)n_t$ are dropped entirely; and government spending becomes the stochastic g_t rather than the fixed g . This backward-looking approximation decouples b_t from the model’s forward-looking variables (π_t, r_t^ℓ) and is what makes a genuinely endogenous, estimable debt stock compatible with the Blanchard-Kahn order/rank conditions – the full nonlinear equation (??) introduces exactly the kind of forward-looking-variable coupling that a naive endogenous-debt extension of this model class runs into. Contrast this with the n_t equation (Subsection 2.7): there, the analogous m_{t-1} and b_{t-1}/b_t terms are *not* dropped, only the fixed-ratio linearization is applied where needed – the difference is that b_t ’s forward-looking coupling comes from r_t/Π_t *multiplying the whole bracket* (including g_t, τ_t), which this equation’s own recursive structure cannot avoid, whereas n_t ’s equation only ever multiplies b_{t-1} and b_t by already-fixed rates, so no comparable simplification is forced on it.

2.4 Euler equation: marginal-utility reformulation (solution technique, not economics)

Paper (eq. ??):

$$\beta E_t \left(\frac{r_{t+1}}{\Pi_{t+1}} \left(\frac{h_{t+1}}{h_t} \right)^\psi \frac{w_t}{w_{t+1}} \frac{A_{t+1}}{A_t} \right) = 1.$$

Estimated model introduces $\mu_t \equiv A_t w_t h_t^\psi$ and rewrites:

$$\mu_t = A_t w_t h_t^\psi, \quad \beta \frac{r_{t+1}^\ell}{\pi_{t+1}} \frac{\mu_{t+1}}{\mu_t} = 1.$$

Algebraically equivalent to the paper's Euler equation, not a behavioral change. The reformulation reduces the count of forward-looking variables in the linearized system (from what the paper's direct specification would imply down to four: π , r^ℓ , μ , y), which is what makes the model numerically tractable for Blanchard-Kahn/E-stability analysis with an endogenous bond stock.

2.5 Labor-leisure condition: productivity in place of the wage, plus the COVID shock

Paper (Section 4.1, combining eqs. ?? and ??):

$$\chi h_t^s + \frac{c_t}{w_t} h_t^\psi = \eta^{-1}.$$

Estimated model:

$$\chi h_t^s + \frac{c_t}{z} h_t^\psi = \frac{1}{\eta} \exp(\varepsilon_t^L).$$

Two changes: the denominator switches from the (endogenous) real wage w_t to the fixed productivity parameter z ; and a new iid shock $\varepsilon_t^L \sim N(0, \sigma_L^2)$ multiplies the right-hand side. A negative realization of ε_t^L shrinks the right-hand side, forcing lower labor supply for given consumption – a negative labor-supply shock, included as a simple stand-in for the COVID-19 labor-supply collapse. (A deterministic one-period dummy, `varexo_det`, was considered first to target a specific calendar quarter, but Dynare 7.1's estimation routine does not read `varexo_det` series from the data file, so it would be silently ignored by the Kalman filter; the iid shock avoids this.) This shock has no counterpart anywhere in the paper.

2.6 New variable: debt-to-GDP link

Not present in the paper at all. A purely definitional auxiliary variable is added to connect the model's debt-to-output state b_t to an observable debt-to-GDP ratio:

$$bT_t = \frac{b_t}{1 + g}.$$

2.7 Reserve equation: restored debt-state coupling (update)

Unlike every other modification catalogued here, this one *reduces* the gap to the paper rather than adding to it. The estimated model's equation for reserves n_t – the analog of the paper's eq. (??), $n_t = m_{t-1}/\Pi_t + b_{t-1}/\Pi_t - b_t/r_t$ – had been implemented as

$$n_t = \frac{m_{t-1}}{\pi_t} + b_0(1 - \omega) \left(\frac{1}{\pi_t} - \frac{1}{r_t^\ell} \right),$$

using the fixed steady-state debt target $b_0(1 - \omega)$ in place of the actual debt state, so n_t never responded to fluctuations in outstanding debt at all. This is now corrected to

$$n_t = \frac{m_{t-1}}{\pi_t} + \frac{b_{t-1}}{\pi_t} - \frac{b_t}{r_t^\ell},$$

the model’s direct, literal analog of eq. (??), with no further simplification needed: b_t ’s own law of motion (the fixed-ratio linearization in Subsection 2.3) never depends on any forward-looking variable, so it remains safe to use here in place of the fixed placeholder without reopening the Blanchard-Kahn problem that motivated that linearization in the first place. Verified directly: the steady state and the Blanchard-Kahn eigenvalue count (4 forward-looking variables) are unchanged before and after this fix.

3 New stochastic structure

Update: the paper’s Section 3 now already includes most of this. At the time this catalog was first written, the paper’s calibrated model (Sections 1–3, 4.1–4.2) had exactly *one* stochastic shock, the preference shock a_t in the utility function. The paper’s Section 3 has since been edited (by instruction, separately from the estimated-model changes tracked elsewhere in this note) so that four more of these processes – z_t , $v_{a,t}$, $x_{a,t}$, and g_t – are now introduced directly in the paper itself, each at the point where the corresponding parameter first appears (productivity z_t in the Firm decision subsection, eq. ??; shopping-time preference $v_{a,t}$ in the Household Problem subsection, eq. ??; deposit technology $x_{a,t}$ in the Banking Sector subsection, eq. ??; government spending g_t in the Policy subsection, eq. ??), using the identical AR(1) functional forms as the estimated model. *Only the three rows marked “still new” below remain absent from the paper:* the monetary-policy shock z_t^r (which would belong in Section 2’s interest-rate rule, eq. ??, not Section 3, and was not touched); the tax-rule shock ε_t^τ (which requires the tax rule’s lag/persistence restructuring described in Subsection 2.2, not made to the paper’s eq. ??); and the COVID labor-supply shock ε_t^L (which has no natural home in the paper’s derivation, since it is not tied to any single fixed parameter the way the other four are).

Process	Law of motion	Status relative to the paper
a_t (preference)	$a_t = a_{t-1}^{\rho_a} \exp(\varepsilon_{a,t})$	Same functional form; original shock in the paper
z_t (productivity)	$z_t = z_{t-1}^{\rho_z} z^{1-\rho_z} \exp(\varepsilon_{z,t})$	Now in the paper (eq. ??)
$v_{a,t}$ (shopping-time pref.)	$v_{a,t} = v_{a,t-1}^{\rho_{va}} v_a^{1-\rho_{va}} \exp(\varepsilon_{va,t})$	Now in the paper (eq. ??)
$x_{a,t}$ (deposit tech.)	$x_{a,t} = x_{a,t-1}^{\rho_{xa}} x_a^{1-\rho_{xa}} \exp(\varepsilon_{xa,t})$	Now in the paper (eq. ??)
g_t (govt. spending)	$g_t = \rho_g g_{t-1} + (1 - \rho_g)g + \varepsilon_{g,t}$	Now in the paper (eq. ??)
z_t^r (MP shock)	$z_t^r = b_r z_{t-1}^r + \varepsilon_t^r$	Still new: no shock in the paper’s rule (??)
ε_t^τ (tax shock)	iid, enters tax rule directly	Still new: rule (??) is deterministic
ε_t^L (COVID labor supply)	iid, enters labor-leisure FOC	Still new: no counterpart in the paper

Six measurement-error shocks (one per non-debt observable) and a seventh for bT^{obs} are added on top of these – see Section 4. These remain entirely absent from the paper, which defines no observable equations at all (Section 4 below).

4 Observable/measurement block (entirely new)

The paper’s Sections 1–4.2 define no mapping from model variables to data at all. The estimated model adds seven measurement equations, each a log-deviation from steady state plus measurement error:

Observable	FRED series	Transformation
y^{obs}	GDPC1 / OPHNFB	$\log(\text{GDPC1}/\text{OPHNFB})$, demeaned
d^{obs}	DPSACBW027SBOG / OPHNFB	$\log(\text{Deposits}/\text{OPHNFB})$, demeaned
π^{obs}	GDPDEF	$\log(\text{GDPDEF}_t/\text{GDPDEF}_{t-1})$, demeaned
r^{obs}	BAA	$\log(1 + \text{BAA}_q/400)$, demeaned
$r^{n,obs}$	FEDFUNDS	$\log((1 + \text{FEDFUNDS}/100)^{0.25})$, demeaned
τ^{obs}	FGRECPT / GDP	$\log(\text{Tax receipts}/\text{Nominal GDP})$, demeaned
bT^{obs}	FYGFDPUN / GDP	$\log((\text{FYGFDPUN}/1000)/\text{GDP})$, demeaned

Two choices are worth flagging as deliberate departures from the most obvious extension of the model: the loan-rate observable uses Moody’s Baa corporate bond yield rather than the bank prime rate, because Baa carries genuine credit-spread variation instead of being mechanically pinned to the federal funds rate plus a fixed spread; and the debt observable uses federal debt *held by the public* (FYGFDPUN), not gross/total public debt (GFDEBTN), divided by nominal GDP (GDP). (An earlier attempt used NGDPDQ as the GDP denominator; that series does not exist on FRED at all, and demeaned the debt ratio in levels rather than logs, inconsistent with the measurement equation above – both fixed in the current build.) A currency-in-circulation observable, present in some earlier drafts of this estimation, is excluded; the model is estimated on seven observables, not eight.

5 Calibration changes

Table 5 compares every parameter given a numerical value in Subsection 4.2 against its estimated-model counterpart. Frequency matters here (Section 1): the paper’s values are not quarterly.

Symbol	Paper (Sec. 4.2)	Estimated model	Status
χ	1.0	0.5 (recalibrated)	Calibrated in both
ψ	1.3	1.3	Unchanged
ν	0.25	0.25	Unchanged
ϕ_n	0.0005	0.0005	Unchanged
ϕ_p	50	50	Unchanged
Ω	0.0	0.0	Unchanged
x_n	0.95	0.95	Unchanged
θ	–	6.0	New (goods demand)
β	0.972	0.9967	Recalibrated for c
v_a	1.138	2.5035 (initial)	Now estimated
η	1.674	1.5294 (initial)	Now estimated
x_a	3.541	5.2528	Still calibrated (c
ρ_a	0.9	0.95	Recalibrated; fixe
g	0.2	0.2	Unchanged
$\bar{\tau}$ (paper: τ)	0.18	0.14 (recalibrated)	Deficit target cha
α_i (paper), a_i (est.)	≈ 0.3 (Table ??)	0.181 (post.; prior mean 0.49)	Now estimated
b_π	(varied in analysis, no fixed value)	3.47 (post.; prior mean 1.00)	Now estimated
φ_b	(varied in analysis, threshold < 0.153)	0.069 (post.; prior mean 0.05)	Now estimated

Parameters with no counterpart in the paper at all, needed only by the extensions above: b_y (output-response weight, new), b_r (monetary-policy-shock persistence, new), $\rho_z, \rho_{xa}, \rho_{va}, \rho_g, \rho_\tau$ (persistence of the newly stochastic processes in Section 3, all absent as concepts from the paper), π^{ss} and $r^{\ell,ss}$ (fixed nominal anchors used only in the linearized debt equation and the Taylor-rule normalization, Section 2.3), and $\underline{r}^n = 1$ (the ZLB floor itself).

6 Parameters moved from calibrated to estimated

Eight structural parameters are estimated via Bayesian methods rather than calibrated: $v_a, \eta, b_\pi, b_y, a_i, b_r, \rho_z, \varphi_b$. Of these, only v_a, η , and (implicitly, via the Section 4.3 stability sweep) φ_b appear as fixed numbers anywhere in the paper’s calibrated model; b_π, b_y, a_i, b_r , and ρ_z either have no fixed calibrated value in the paper (they are swept over ranges for the determinacy analysis, or $-b_y, b_r$ – do not exist in the paper’s model at all).

Prior revision (current round). b_π and b_y ’s priors were substantially loosened – from Normal(2.10, 0.15) and Gamma(0.28, 0.10) to Gamma(1.00, 1.00) and Gamma(1.00, 0.70) – so that the posterior on the Taylor-rule inflation-response coefficient is not mechanically pulled toward the earlier round’s own tight prior; φ_b ’s prior mean was also lowered, from 0.10 to 0.05. Priors and current-round posterior results:

Symbol	Prior	Prior mean/std	Post. mean (90% HPD)
v_a	Gamma	2.50 / 0.75	2.890 [2.055, 3.482]
η	Gamma	1.53 / 0.40	1.136 [0.922, 1.326]
b_π	Gamma	1.00 / 1.00	3.474 [3.344, 3.604]
b_y	Gamma	1.00 / 0.70	0.083 [0.057, 0.106]
a_i	Beta	0.49 / 0.10	0.181 [0.169, 0.195]
b_r	Beta	0.36 / 0.10	0.262 [0.171, 0.347]
ρ_z	Beta	0.45 / 0.10	0.443 [0.292, 0.549]
φ_b	Beta	0.05 / 0.05	0.069 [0.030, 0.101]

With the informative prior removed, b_π ’s posterior does not fall toward the diffuse prior’s mean of 1 – it moves to 3.47, *higher* than the earlier tight-prior estimate of 2.17. The data want a stronger inflation response than the previous prior allowed; the conclusion that the Taylor principle holds is a property of the data, not an artifact of prior choice. b_y ’s prior needed reshaping once during this process: an initial Gamma(1.00, 1.00) (shape 1, i.e. Exponential, whose density is maximized at zero) put b_y ’s posterior mode at exactly 0.0000 with a non-positive-definite Hessian there – a corner solution the prior itself pre-loads. Raising the shape via Gamma(1.00, 0.70) (mode near 0.51, no boundary pre-loading) resolved this without narrowing the prior’s effective diffuseness. a_i is essentially unchanged from the previous round (0.181 vs. 0.180) despite its own prior being untouched, a useful cross-round consistency check.

ρ_a and ρ_{xa} were also tried as free parameters in an earlier attempt; freeing them drove ρ_a to ≈ 0.9992 – effectively a unit root – and produced a non-positive-definite Hessian at the posterior mode, so both remain fixed at 0.95 in all reported results.

7 Estimation methodology (absent from Sections 1–4.2 by construction)

Sections 1–3 and Subsections 4.1–4.2 describe only a calibrated, deterministic-steady-state model; nothing about Bayesian inference appears until Section 5 (which, as noted in the abstract, is itself outdated relative to the results actually obtained). Briefly, for a replacement Section 5 to describe accurately:

- **Two-stage mode-finding.** OccBin’s piecewise-linear likelihood is nonsmooth at ZLB regime switches, so the posterior mode is first located on the linear (non-OccBin) approximation (`mode_compute=4`), then MCMC is restarted at `mode_compute=0` from that mode with the true OccBin likelihood switched on.
- **MCMC jumping covariance.** An initial run using a diagonal proposal covariance (`prior_variance`) produced chain disagreement traceable to that covariance ignoring the b_π/b_y correlation. A Hessian-based covariance fixed this under the original calibration but showed renewed (milder) disagreement after recalibrating $\bar{\tau}, \chi$; that round’s reported (“gen1”) results instead used a jumping covariance built from a preliminary round’s own pooled posterior draws – standard adaptive-MCMC pilot/tune practice, verified against Dynare’s source to get a subtle sign/inversion convention right (the saved matrix must be a precision, not a covariance). The current, diffuse-prior round did not need this extra step: a plain Hessian-based covariance (evaluated at the Stage-1 mode) was sufficient throughout once the jump scale was tuned down, and achieved the best chain-mean agreement of any round for three of the four policy parameters.
- **Acceptance-ratio discipline.** Any chain with an acceptance ratio below 0.05 is treated as a failed run and re-estimated with an adjusted jump scale; short (3,000-draw) diagnostics systematically overstated what a full 50,000-draw run would sustain, so later rounds built in extra margin before committing to a full run. The current round’s first production attempt cleared the 0.05 floor (5.4%/5.3%) but only barely, with weak Geweke convergence – rather than accept a technically-passing but marginal result, it was treated as a failure, and a longer (8,000-draw) diagnostic was used before the next full run, since the usual 3,000-draw diagnostic had just mispredicted that round’s production behavior especially badly ($\sim 5\times$ degradation, worse than any earlier round).
- **Smoother limitation.** OccBin’s nonlinear smoother did not reach a fixed point in any round attempted, so no OccBin-consistent smoothed shock/state decomposition is available; this does not affect the posterior parameter estimates, which are informed by OccBin’s *filter* (used throughout MCMC) rather than the smoother.

The full tuning history, Geweke convergence diagnostics, determinacy/ E-stability results at the estimated posterior, and steady-state implications are documented in `mmt8b0ccBin3.covid_bT.baa_epsL.summ` and are not reproduced here.